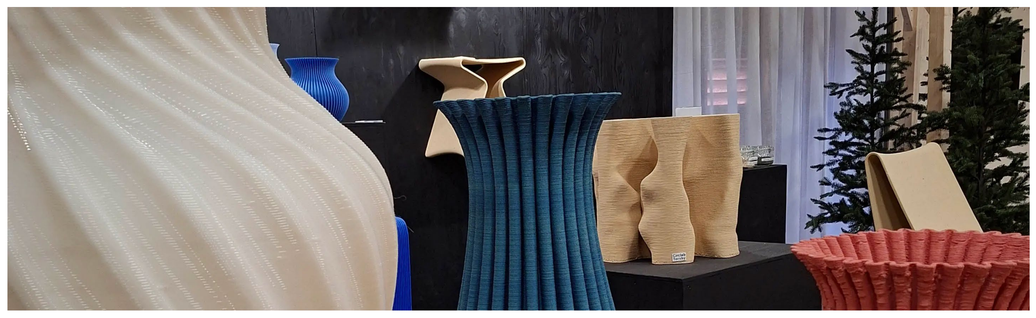
3D printing with thermoplastic and biobased composites at an industrial testbed as an authentic resource in teaching of the design process

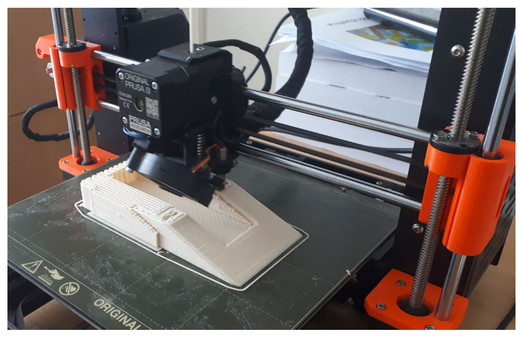
Robin Strandberg[1,2] Magnus Lestelius[1], Anders Biel[1] & Jesper Haglund[1]

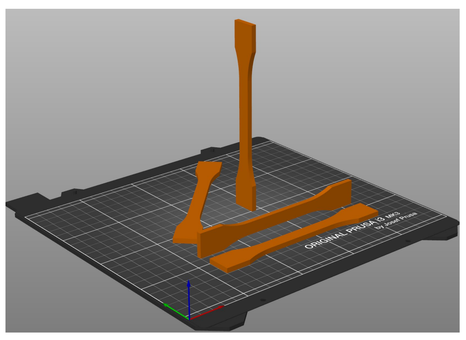
1. Karlstad University, Sweden, 2. Stjerneskolan, Torsby, Sweden

Context of the study

In our research project, a collaboration has been formed between upper-secondary school teachers in the chemistry and technology subjects at Stjerneskolan, researchers in science and technology education at Karlstad University, and Circlab, an industrial testbed for sustainable additive manufacturing, by 3D printing of composites from thermoplastics, bio-based and recycled materials.







Resources at the 3D printing testbed

The facilities at Circlab are available on a continuous basis for participating teachers and pupils. This includes access to up-to-date equipment for CAD modelling, 3D printing, 3D printable materials (composites) and mechanical testing of artefacts.

In this way, the testbed provides opportunities for increased authenticity in the teaching of the design process, with connections to the demands of present customers, and involvement in construction of prototypes and finished products.

References

Barab, S., & Squire, K. (2004). Design-based research: Putting a stake in the ground. *Journal of the Learning Sciences, 13*(1), 1-14.

Chien, Y.-H. (2017). Developing a pre-engineering curriculum for 3D printing skills for high school technology education. *Eurasia Journal of Mathematics, Science and Technology Education, 13*(7), 2941-2958.

Ford, S., & Minshall, T. (2019). Invited review article: Where and how 3D printing is used in teaching and education. *Additive Manufacturing, 25*, 131-150.

Study design

The present study is conducted as a design research study in teaching of technology at the upper-secondary technology programme, in which teaching sequences are developed, tested, and evaluated through rounds of iterations.

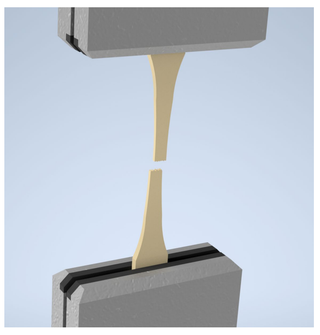
Video observation of pupils' interaction with the resources at the testbed and amongst themselves, and group interviews with pupils about their experiences will be conducted.

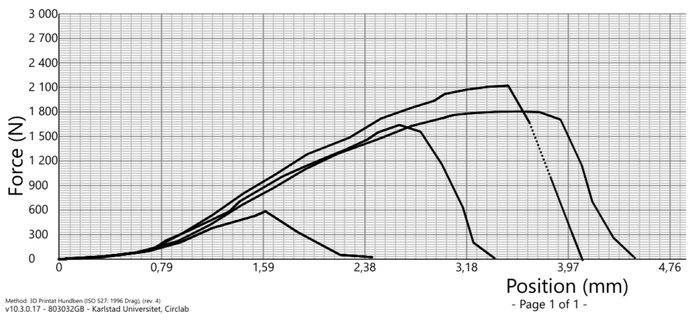
Teaching design

The action research will be conducted as two modules.

In the first module, pupils will be introduced to 3D scanning of objects and 3D printing of models of the objects in different materials.

In the second, more advanced module, pupils will draw a standard object in CAD, a "dogbone", 3D print it in different materials and along different printing directions, and make a tensile test of its material properties.





Expected outcomes and implications

As a planned outcome of the study, teaching sequences will be developed and tested, in which pupils gain insight into the design process, composite material properties, and some considerations of sustainability, by access to up-to-date equipment at an industrial testbed.

At the organisational level, implications will be drawn with regard to experiences of a collaboration between a school, an industry, and academia.

Contacts

robin.strandberg@torsby.se

jesper.haglund@kau.se